5-MINUTE REVISION

Missing Plot Analysis

From the chapter's own revision points, formulae and questions.

BOOK-ONLY EDITION · Verified against the textbook

Selected from Chapter 23 · Chapter Summary in the textbook.

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Quick revision · 1

- Record why an observation is missing or unusable.
- A poor response caused by a treatment must not automatically be discarded.
- A substituted value is an estimate, not a recovered observation.
- The CRD can usually be fitted directly with unequal treatment sample sizes.
- A missing plot destroys the complete balance of an RCBD or Latin square.

Quick revision · 2

- With one missing plot, the usual complete-design error degrees of freedom decrease by one, provided the model remains estimable.
- Substitution minimizes error SS; treatment SS and comparison standard errors require the stated adjustments.
- Fitting the model to observed responses is an alternative to substitution.
- Missingness related to an unobserved response can bias the analysis; a calculation alone does not resolve this problem.

Important formulae

For a CRD, analyse the observed responses using their actual treatment sample sizes. If all v treatments remain represented:

$$edf = n_{obs} - v, \qquad SE(\bar{Y}_i - \bar{Y}_j) = \sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

For one missing observation in an RCBD with v treatments and r blocks:

$$\hat{x} = \frac{rB + vT - G}{(r-1)(v-1)}, \qquad edf = (r-1)(v-1) - 1$$

Here B and T are the observed totals of the affected block and treatment, and G is the observed grand total.

For one missing observation in a Latin square of order v:

$$\hat{x} = \frac{v(R + C + T) - 2G}{(v-1)(v-2)}, \qquad edf = (v-1)(v-2) - 1$$

Here R, C and T are the observed totals of the affected row, column and treatment. In both substitution formulae, all totals exclude the missing value. Use the bias corrections and comparison formulae in the chapter; do not treat the substituted value as an independent observation.

Self-check from the chapter

1. The classical substitution estimate of a missing plot minimizes the error sum of _____. **[Squares]**
2. Estimating one missing observation reduces the error degrees of freedom by _____. **[One]**
3. An estimated missing value is not an additional independent _____. **[Observation]**
4. In the RBD missing-value formula, totals exclude the missing _____. **[Value]**